

Michelle Zbizika

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Education

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT)

B.S. in AI and Decision Making (6-4)

Cambridge, MA

2021 - 2025

Relevant Coursework: Design and Analysis of Algorithms; Probability and Random Variables; Theory of Computation; Fundamentals of Programming; Introduction to Machine Learning; Linear Algebra; Differential Equations; Real Analysis; Intro to Deep Learning; Web Lab; Multivariable Calculus;

Great Neck North High School

GPA: 95.5, SAT: 1600

Great Neck, NY

June 2021

Experience

Vatic Investments

August 2025 - Present

Quantitative Researcher

- Designed and built end-to-end a multi-agent autoresearch framework (orchestrator, proposer, judge agents) coordinating through persistent research ledger with gated evaluation funnels that automated signal discovery and post-trade analysis
- Automating hypothesis, generation, evaluation, and documentation research loop previously done by researchers, built post-trade and post-simulation analysis tooling for teams standard research workflow
- Built and deployed a Slack-resident conversational research agent (Anthropic API) used by the research team for querying results and monitoring experiments
- Developed and trained custom deep learning architectures for financial time series on a shared training platform, including a channel-attention (inverted) transformer, classification and pretraining transformer variants, and MLP baselines (PyTorch)
- Scaled model training to multi-node, multi-GPU infrastructure, including DDP/FSDP
- Mentored 2 quantitative research interns, guiding project scoping, code review, and research methodology

Quantitative Research intern, Summer 2024

- Developed and productionized systematic trading signals and execution strategies across futures and equity products.
- Designed and implemented agentic AI systems and foundational models for financial research, market analysis, and trading applications.
- Collaborated in teams with other experienced researchers and Google AI consultants

Take Mobi

Summer 2022

Data Science Intern

- Built machine learning based data analysis pipeline; Spearheaded development of ML models and pipelines for AMEX Trip Planner within the Mobi data science research team
- Applied Graph Neural Networks, RNN, sequential and clustering models on customer-item interaction data using PyTorch and CUDA, generating real time POI Category, and Context recommendations for a feature on AMEX Trip planner deployed models with AWS cloud services, including S3 and sagemaker
- Generated POI prediction accuracy of 78%.

MIT CSAIL Geometric Data Processing Group

Jan - May 2022

Undergraduate Researcher

- Developing Deep Neural Network models using Fourier transform models to advance ML driven image rendering and computer vision
- Implemented neural network models in Pytorch to generate image pixel prediction framework and render higher resolution images

SSP: Summer Science Program in Astrophysics

July 2020

Researcher

- Calculated the orbit of asteroid 1994 LY through telescope and image analysis using astronomy, calculus, linear algebra, matrices, statistics; utilized software including Python, DS9, AstroImageJ, Stellarium
- Published report findings in International Astronomy Union Minor Planet Center

Projects

Robotics Science and Systems MIT 6.4200 Spring 2024

- Programmed full autonomy stack (perception, localization, planning, control) for a small autonomous vehicle; **1st place** in final competition
Built the full autonomy stack for an autonomous racecar — LiDAR-based perception, particle-filter (Monte Carlo) localization, pure pursuit path tracking, and computer-vision lane/object detection

Awards

USA Physics Olympiad (USAPHO) *Semifinalist*: highest scoring woman in NY state; **USA Computing Olympiad Silver Division**: awarded to 1600 pre-college participants, top 20% of all competitors; **American Invitational Math Exam (AIME)** *Qualifier*; **US Earth Science Olympiad** *9th place National*; **Long Island Science Engineering Fair (LISEF)** Semifinalist, Honorable Mention

Leadership & Interests

MIT Varsity Women's Lightweight Crew

Fall 2022 - 2023

Alpha Chi Omega *Social chair*: Organized social events with other organizations for 100+ members in the chapter

Spring 2022 - Winter 2023

MIT Sloan Hackalytics: Participated in hackathon for using data analytics to predict pre-diabetes

Spring 2022

Discover Ocean Engineering: Built remote controlled underwater robots in a 5-day program in collaboration with 2 other students

Fall 2021

Skills: Python, PyTorch, Tensorflow, Pandas, Typescript, Google Cloud Platform, AWS, Machine learning, Mathematical analysis, Data processing

Interests: Performed at Carnegie Hall on Accordion and Violin, Aquarium Cultivation, Figure Skating, EDM music